Investigating Genetic and Biological Differences between African American and Caucasian Endometrial Cancer Patients

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ENDOMETRIAL CANCER

- **65,950** U.S. women predicted to be diagnosed in 2022 alone

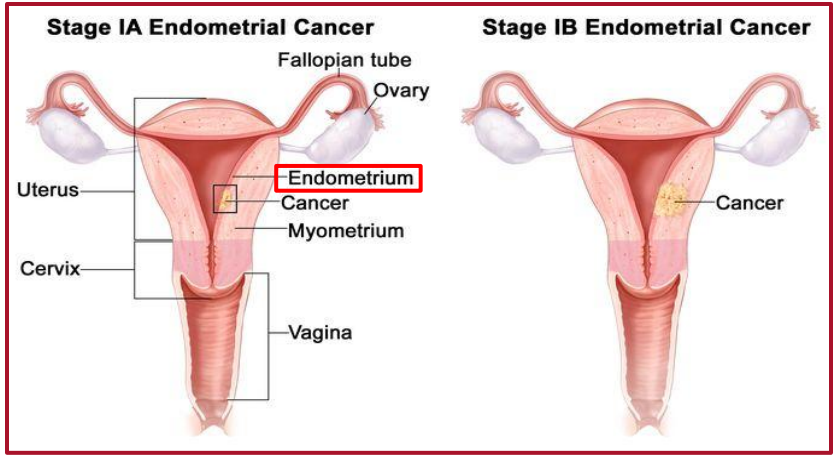


Fig. 1. *Diagram of a Uterus in Stage 1A and 1B of Endometrial Cancer (Winslow, 2021)*

Type 1	Type 2
Estrogen Related	Estrogen **Un**related
Clinically **Indolent**	Clinically **Aggressive**
Low and High Grade Endometrioid Carcinoma **(EC)**	Serous Carcinoma **(SC)** and Mixed Endometrioid Carcinoma **(MEC)**

GENOMIC PREDISPOSITION

- **90% higher 5 year mortality risk** for African Americans (AA) than Caucasians (CS)

Incidence of Gynecologic Cancers in U.S. African American Females

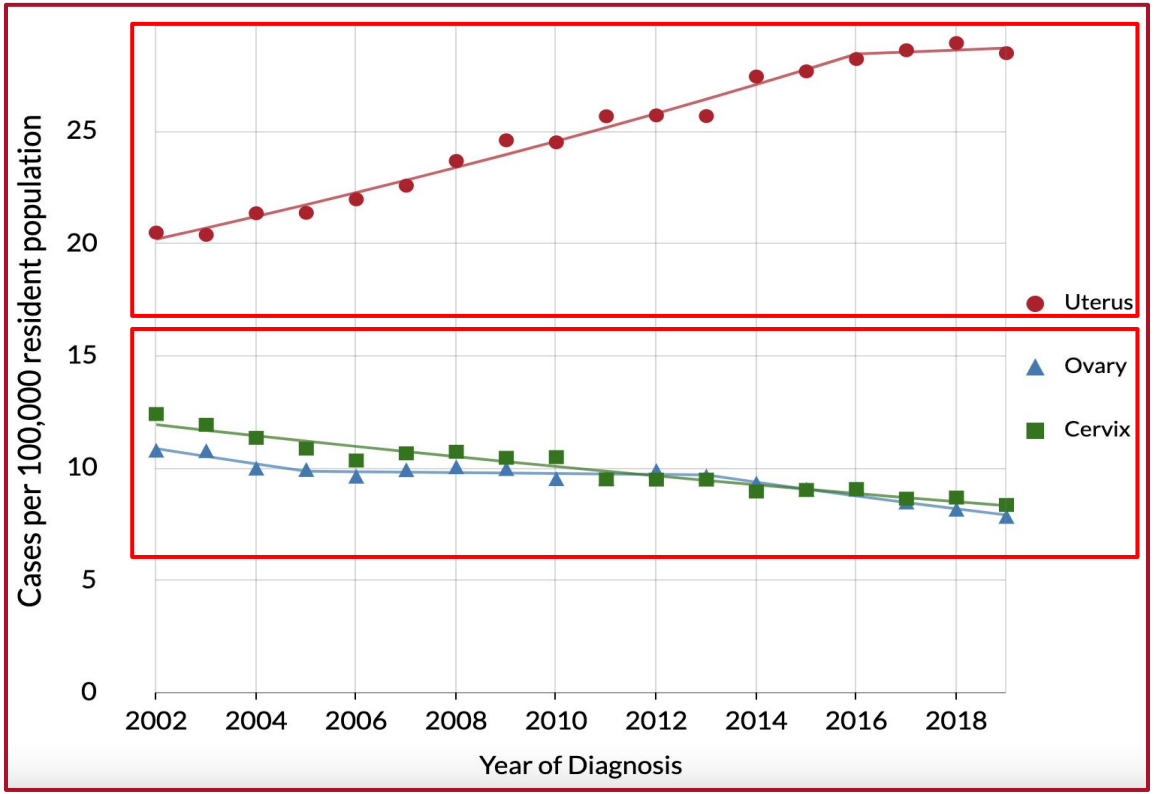


Fig. 2. *Endometrial Cancer Incidence of African American Women in the United States (National Program of Cancer Registries, 2022)*

CURRENT DIAGNOSTICS

Prognosis	Mutation
Good	PIK3CA, ARID1A, & MUC16
Poor	TP53 and PIK3R1

- Little is known about the **mutational landscape** of EC between **racial groups**

PROJECT GOALS

1. **Identify** significant genetic biomarkers based on race of EC patients
2. Evaluate **significant differences** in microbiome signatures, genetic mutations, and protein expression disparities based on **race** across **prognostic groups**

METHODOLOGY

443 de identified profiles from cBioPortal stratified by race

African American

Caucasian

Good prognosis

Poor prognosis

Polymerase Epsilon Ultra Mutated (POLE)

- **Low mutation rates**

Microsatellite Instability (MSI)

- **Mismatch repair deficiency**

Copy Number Low

(No specific molecular profile)

- **Microsatellite stable**

Copy Number High

(p53 mutated)

- **Composed of type 2 cases**

Evaluate **significant differences** in

- Mutational Frequency
- Protein Expression
- Microbial Profiles

PATIENT DEMOGRAPHICS

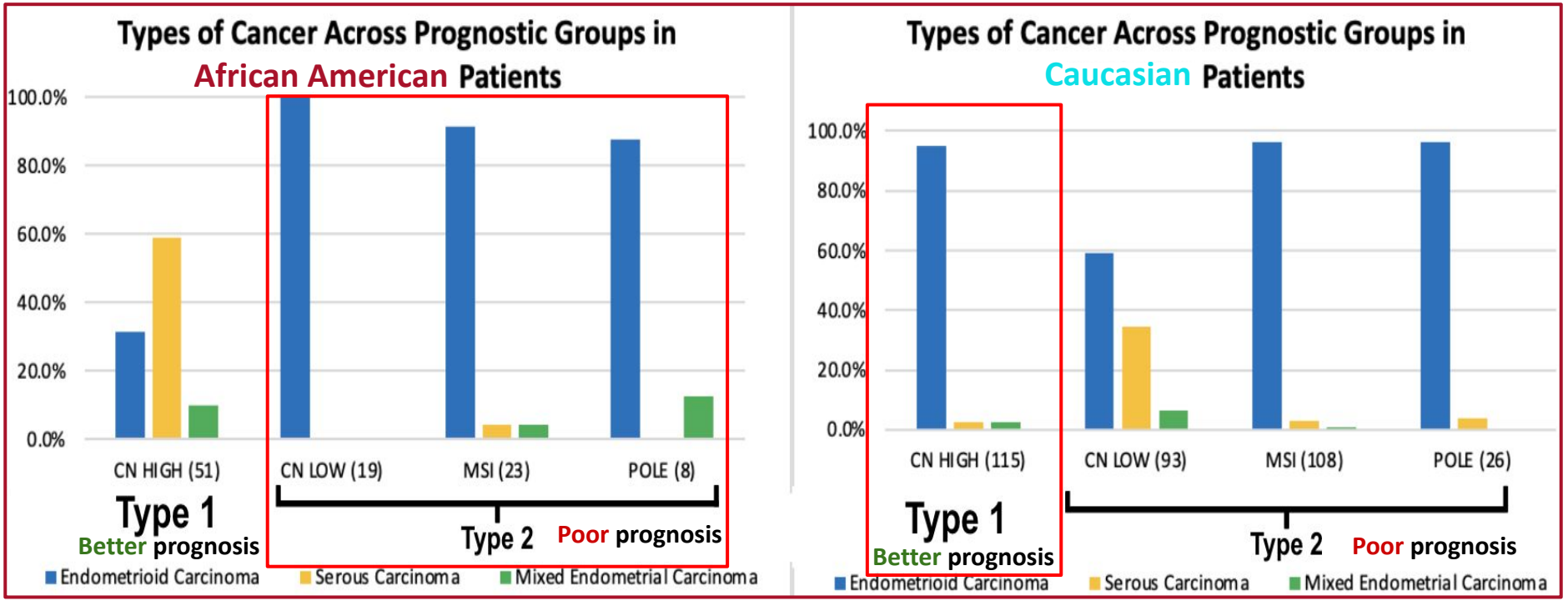


Fig. 3. Types of Endometrial Cancer by Race and TCGA Prognostic Group *AA and CS data were stratified by TCGA prognostic groups (Student generated figures)*

MICROBIOME SIGNATURE

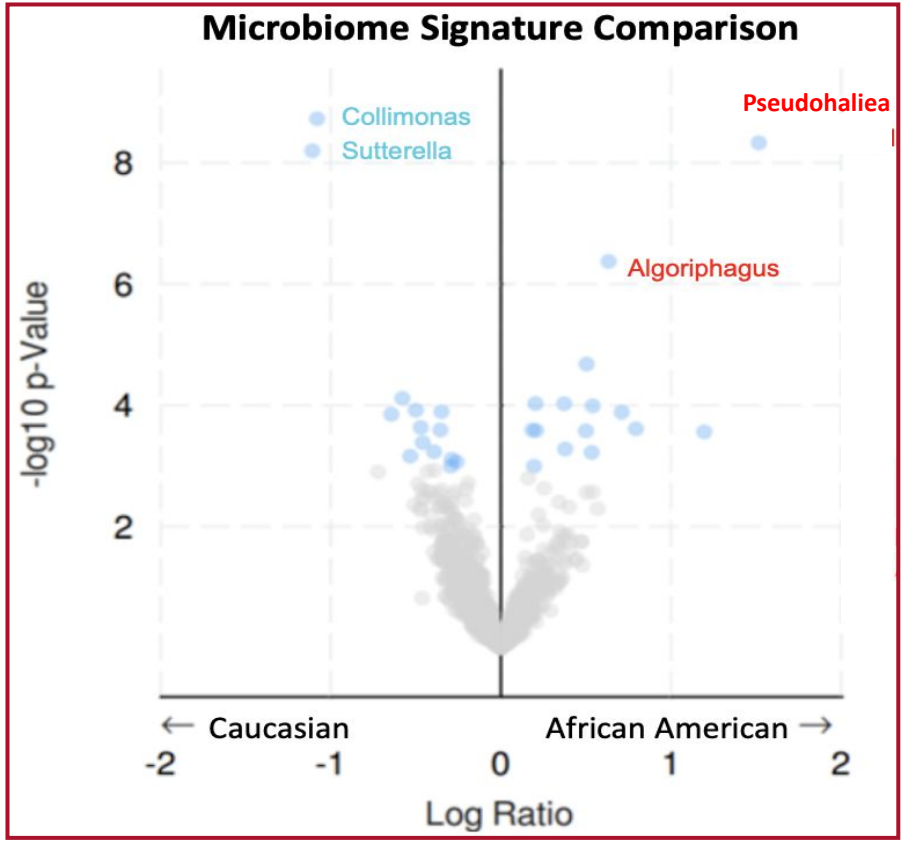


Fig. 4. AA and CS Patient Microbiome Signature Comparison *(p<0.001)*
(Student generated figure)

Significant Microbial Signatures in Caucasians	Significant Microbial Signatures in African Americans
Collimonas → **good** prognosis in breast cancer (Wu, 2022)	**Algoriphagus** → **overweight** EC patients (Zhao, 2022)
Sutterella → **development** of adenomas (Hale, 2016)	**Pseudohaliea** → **not** highly researched

PROTEIN EXPRESSION

Underexpression of MYH11 in African Americans → poor prognosis in lung cancer
(Nie, 2020)

Overexpression of RAD51 in African Americans → increased drug resistance in breast & pancreatic cancer
(Nagathihalli, 2011) (Li, 2021)

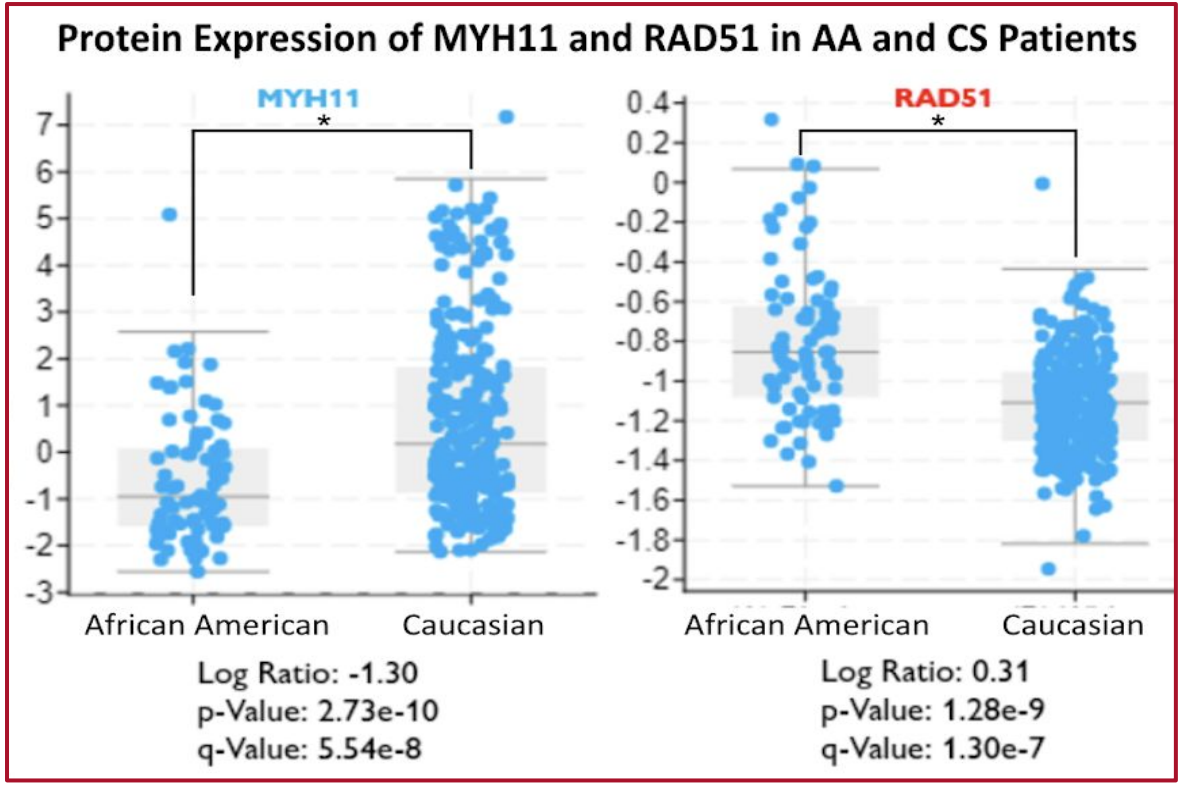
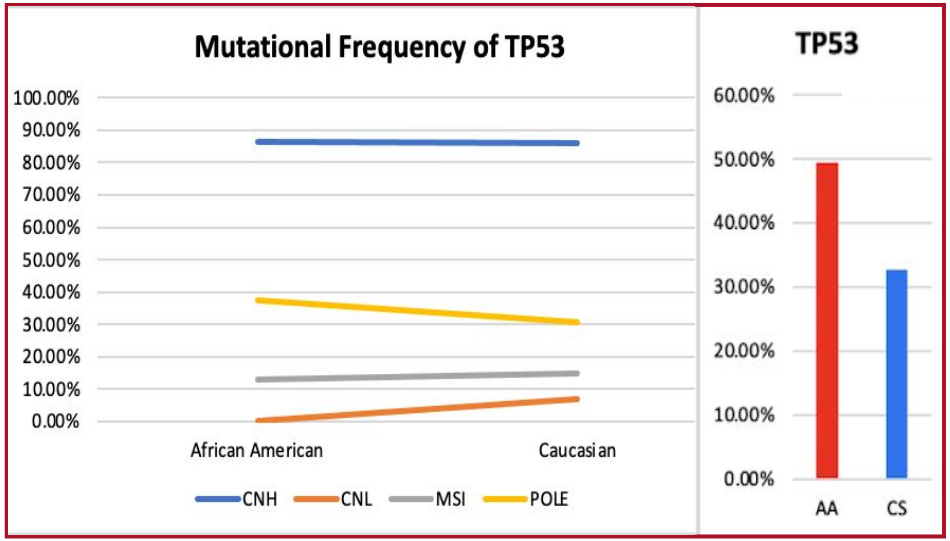


Fig. 5. Protein Expression of MYH11 and RAD51 in AA and CS Patients *MYH11 expression was higher in CS patients while RAD51 expression was higher in AA patients. (p<0.001)* ***(Student generated figures)***

SIG. HIGHLY MUTATED GENES IN AFRICAN AMERICANS



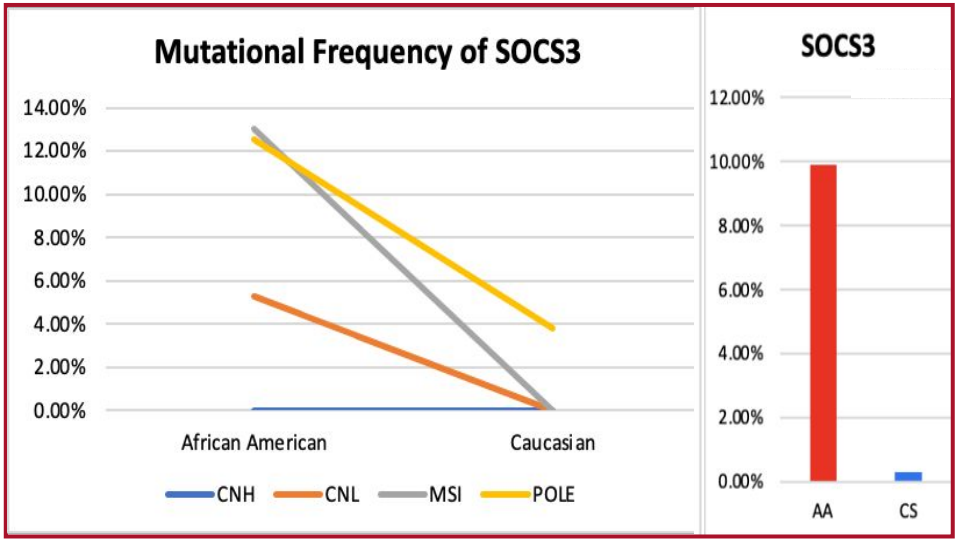


Fig. 7. Mutational Frequency Across TCGA Prognostic Groups *TP53 and SOCS3 ($p<0.001$) (Student generated figures) (Y-axis scales are different between graphs)*

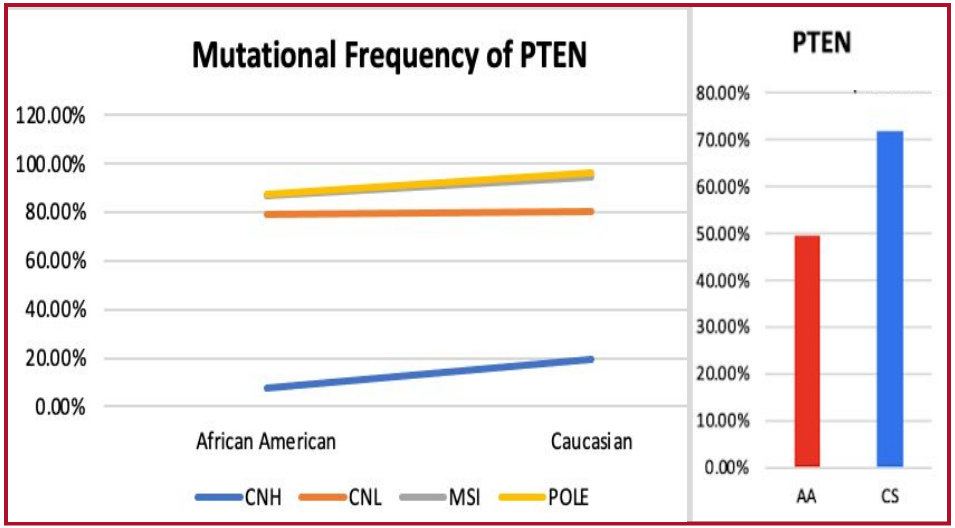
TP53

- Sig. more prevalent in **AA** & type **2** patients (**poor** prognosis)

SOCS3

- Sig. more frequent in **AA**
- Mostly **absent** in **Caucasians**
- Role in carcinogenesis (Chu, 2018)

SIG. HIGHLY MUTATED GENES IN CAUCASIANS



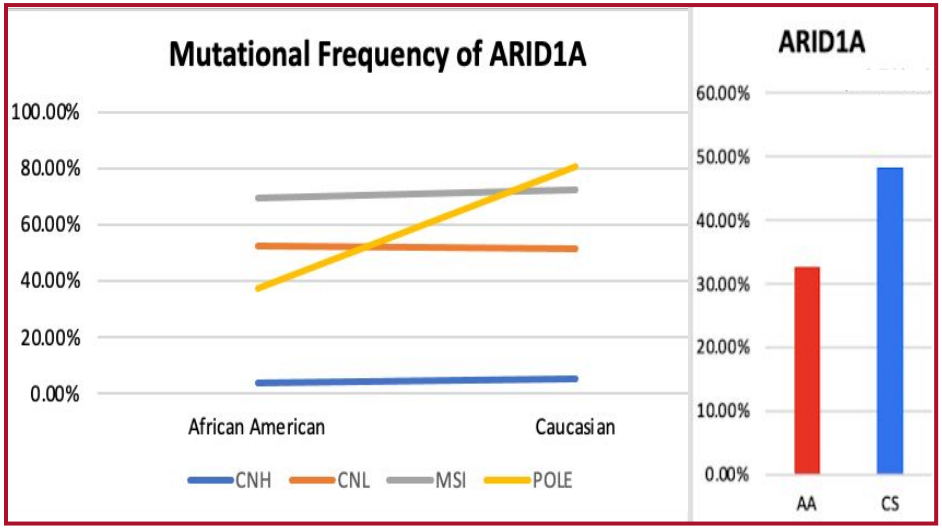


Fig. 6. Mutational Frequency Across TCGA Prognostic Groups *PTEN and ARID1A (p<0.001) (Student generated figures) (Y-axis scales are different between graphs)*

PTEN

- Sig. more frequent in **Caucasians** & type **1** patients (**favorable** prognosis)

ARID1A

- Sig. more frequent in **Caucasians**
- Underexpression → **poor prognosis** in breast cancer (Guan, 2011)

LIMITATIONS

- Disproportionately lower **POLE** cases (good prognosis and high mutation rates)
- **Caucasian** patients were **3 fold more** than **African Americans**
- Geographic and environmental limitations

CONCLUSIONS

Significant Biomarkers for African Americans	Significant Biomarkers for Caucasians
Overexpression of **RAD51** Proteins	Overexpression of **MYH11** Proteins
TP53 and **SOCS3** mutations	**PTEN** and **ARID1A** mutations

Higher mortality rates in African Americans

Lower mortality rates in Caucasians

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